Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name: HackSmith

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Problem statement: Safer AI-assisted network incident repair

Brief about the idea

TwinOps

AI proposes a network repair. A digital twin tests it.
An engineer approves the exact change.

During an outage, a fix can restore one connection while breaking another.
TwinOps makes that trade-off visible before a change reaches the lab network.

For network operations engineers, SREs and change approvers.
Theme: Future of Work & Enterprise Productivity.

Opportunities

What differentiates it

A repair proposal includes reproduced symptoms, rejected alternatives, challenge results and rollback evidence that an approver can inspect.

How it solves the problem

Competing fixes run on an isolated protocol twin. The recorded broad rollback restored the initial journeys but failed a later IPv6 challenge.

Unique value proposition

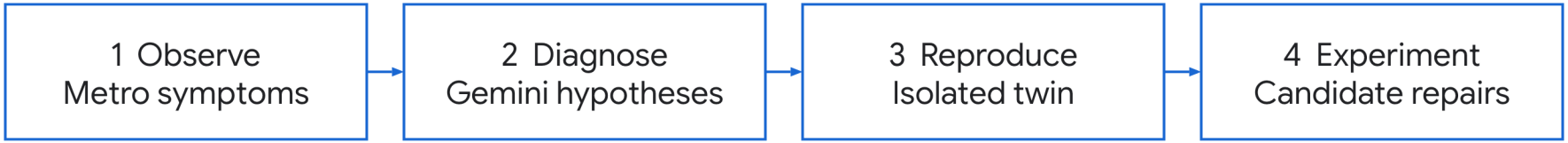
Approval is bound to the tested change and target state. Fresh checks verify the outcome; missing evidence can stop the workflow.

List of features offered by the solution

Incident detection	Reads Metro observations without receiving the selected fault label.
Gemini investigation	Ranks hypotheses and proposes typed, allow-listed repairs.
Twin experiments	Tests real routing, DNS and TLS in a separate network.
Adversarial checks	Challenges candidate repairs and retests bounded revisions.
Human decision	Shows the exact diff, evidence limits and rehearsed rollback.
Verified outcome	Checks recovery after apply and exports the Proof Pack.

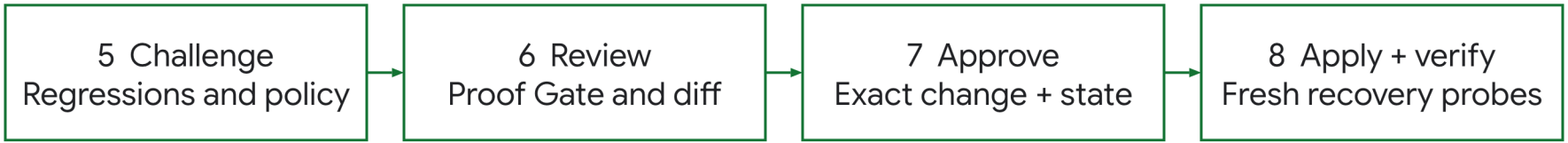
Process flow diagram

Observation and experiments



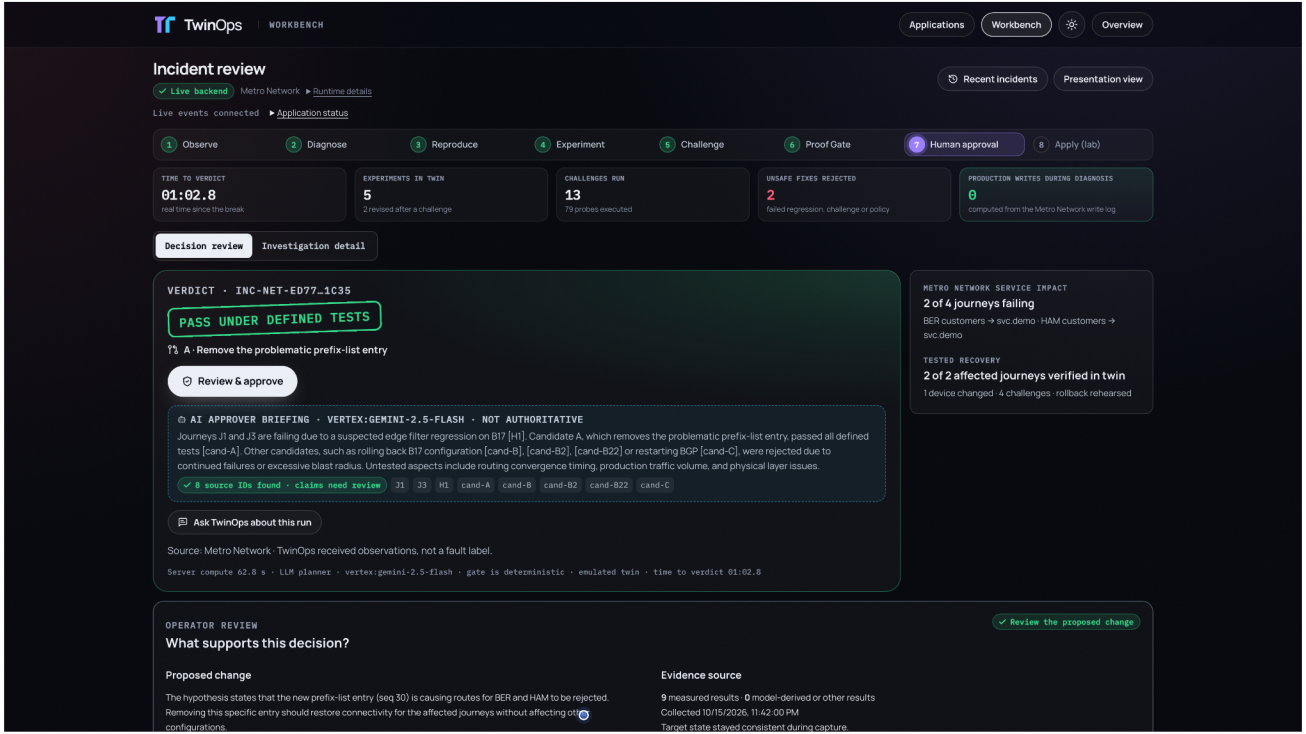
After experiments, challenge each eligible repair before review.

Evidence and authorized change



Insufficient evidence stops approval. A failed apply or recovery check triggers rollback handling.

Wireframes / interface: the implemented workbench



Decision review

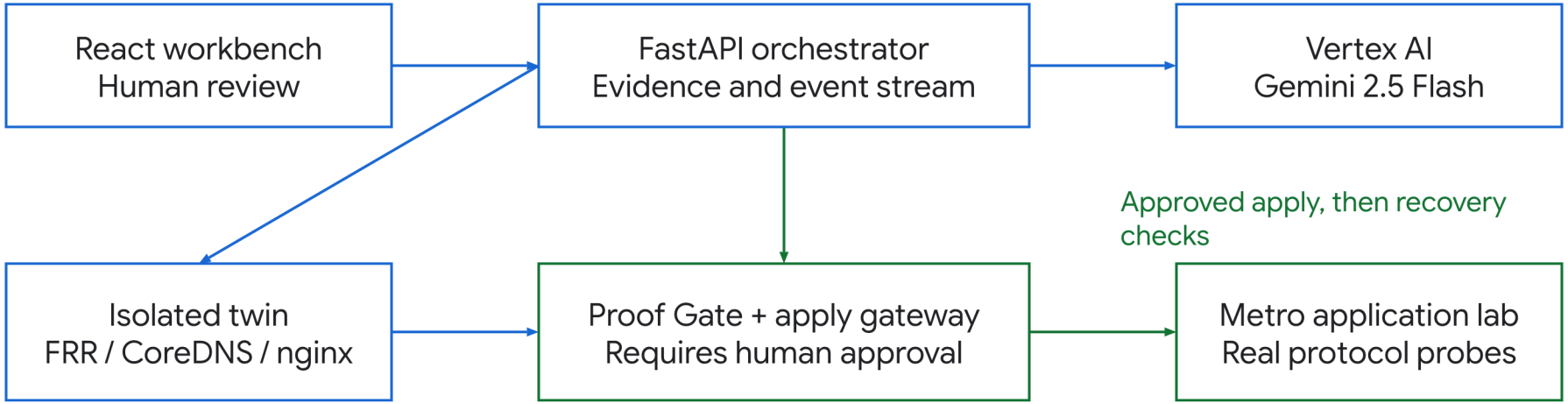
The operator sees the selected repair, remaining impact and supporting tests.

Approval is still pending here.

The passing twin result alone does not change Metro.

Architecture diagram of the proposed solution

Application and lab networks on Cloud Run. Model access through Vertex AI.



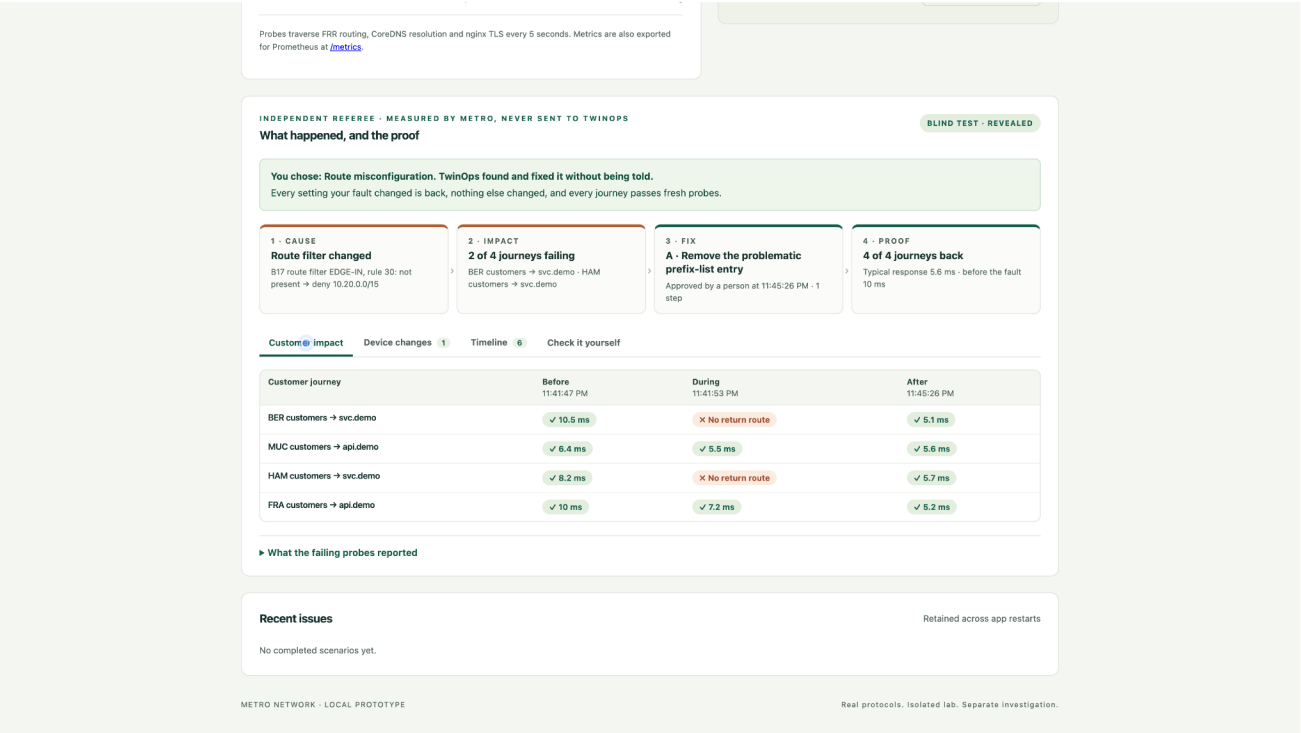
SQLite journals evidence on the instance. Secret Manager supplies the approval credential.
Cloud Build and Artifact Registry build and store the deployment image.

Technologies used in the solution

Technology	Role in the working prototype
Cloud Run (second generation)	Hosts TwinOps, Metro and isolated Linux namespace networks
Gemini 2.5 Flash / Vertex AI	Hypotheses, repair proposals, challenges and bounded revisions
Google Gen AI SDK	Structured model requests with schema and catalogue validation
Secret Manager	Provides the private operator approval credential
Cloud Build / Artifact Registry	Builds and stores the deployment container
React / FastAPI / SQLite	Review interface, orchestration and instance-local evidence
FRR / CoreDNS / nginx	Real routing, DNS, IPv4/IPv6 and TLS experiments

Firebase and Firestore are not used. Cloud Run provides the deployment path.

Snapshots of the prototype



Measured recovery

Berlin and Hamburg failed during the fault.

One approved rule removal restored all four journeys.

Metro measures the outcome separately from the planner.

Prototype performance report / benchmarking

Evidence scope	Observed result
Hosted Gemini repair 15 October 2026	10 model calls, 0 fallbacks and 0 diagnosis writes
Candidate experiments Same hosted repair	5 experiments, 13 challenges and 79 probes
Post-change recovery Same hosted repair	11/11 checks passed in each of 3 observation windows
Deterministic evaluation 13 October 2026	320 cases: 200 verified repairs, 120 abstentions, 0 unsafe passes; 13/13 controls passed

The 320-case suite uses the runbook/model twin and makes zero Gemini calls.
One hosted repair does not establish production reliability or measured time savings.

Additional details / future development

Today: a shared prototype

Routing, DNS and TLS exercises on synthetic topology.
One lab instance, ephemeral evidence and a shared approval credential.

Next: operator evaluation

Compare investigation and review effort against a baseline.
Measure escalation, rollback outcomes and decision quality.

Then: a read-only shadow pilot

Validate network adapters and twin coverage against real environments before enabling operational change.

For wider deployment

Durable evidence storage, individual identity, per-target isolation, owned monitoring and tested recovery are required.

Public repository, demo video and final product

GitHub public repository

https://github.com/prabhakarcs786/twinops

Demo video

https://youtu.be/0LjDHiUBy0Y

Final product: TwinOps workbench

https://twinops-live-705406130795.europe-west1.run.app/workbench

Connected application: Metro Network

https://twinops-live-705406130795.europe-west1.run.app/metro/

Review access and walkthrough: GitHub / docs / JUDGE_GUIDE.md

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Thank you

